Antonin Corinaldi

2nd year PhD student in cosmology

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A. Corinaldi antonincorinaldi Antonin Corinaldi

Education

PhD in Astronomy and Astrophysics *2025 – 2028*
Université Paris Cité

Post-master research year (post-master professional gap year) *2024 – 2025*
Université Paris-Saclay

Master's degree (M2) in Astronomy and Astrophysics *2023 – 2024*
Observatoire de Paris

- Final grade : 15.5/20 ; final ranking : 10/33

Magistère in Fundamental Physics, Orsay *2021 – 2024*
Université Paris-Saclay

Scientific preparatory classes for Grandes Écoles, PCSI / PC* *2019 – 2021*
Lycée Michelet – Vanves

Scientific Baccalauréat, highest honors (final grade : 18.71/20) *2019*
Lycée Montesquieu – Le Mans

Research Experiences

PhD thesis in cosmology *2025 – 2028*
CEA Astrophysics Department, CosmoStat team, Paris-Saclay

- **Supervisors** : Martin Kilbinger and Sandrine Codis
- **Name of the thesis** : *"Uncovering the 3D cosmological tidal field of dark matter"*

Post-master research internship (3 months of preparation and 6 official months) *2024 – 2025*
CEA Astrophysics Department, CosmoStat team, Paris-Saclay

- **Supervisors** : Martin Kilbinger and Calum Murray
- **Topic** : cosmology and galaxy survey science, 3D measurement of galaxy intrinsic alignment with UNIONS and Euclid, data analysis, simulation-based inference, Python programming
- Member of the **UNIONS** and **Euclid** collaborations (member of Euclid's "intrinsic alignment" Key Project)

M2 research internship (4 months) *2024*
Institut de Recherche en Astrophysique et Planétologie (IRAP), Toulouse

- **Supervisor** : Isaac Tutusaus
- **Topic** : cosmology and galaxy survey science, forecasts on cosmological constraints from Euclid's future photometric galaxy surveys using 3x2pt analysis (statistical inference with weak lensing, galaxy clustering, and cross-correlations), study of the impact of photometric redshift measurement errors on cosmology, Python programming

M2 numerical project (15 sessions of 3 hours) *2023 – 2024*
Institut d'Astrophysique Spatiale (IAS), Orsay

- **Supervisor** : Marian Douspis
- **Topic** : cosmology, characterization of galaxy cluster catalogues detected via the Sunyaev–Zel'dovich effect using artificial intelligence (artificial and convolutional neural networks), Python programming

M1 research internship (3 months) *2023*
Royal Observatory of Edinburgh, Scotland

- **Supervisor** : Yan-Chuan Cai (co-supervisor : Stéphane Ilic)
- **Topic** : cosmology, creation of a numerical model for cosmic supervoid formation and theoretical prediction of the associated integrated Sachs–Wolfe effect, Python programming
- Recipient of the **IDEX excellence scholarship**

L3 research internship (6 weeks) *2022*
Laboratoire Irène Joliot-Curie (IJCLab), Orsay

- **Supervisor** : Stéphane Ilic
- **Topic** : cosmology, analysis of cosmic microwave background maps and study of the impact of the epoch of reionization on constraints of the reionization parameter τ for Simons Observatory and CMB-S4 (statistical inference), Python programming

Skills

Programming languages : Python, C, C++, Fortran, HTML

Astrophysics codes: *CosmoSIS* pipeline (M2 internship), *Class* numerical code (L3 internship)

Other digital tools: *GitHub* environment, *LaTeX* (scientific report writing during L3, M1, M2 internships and projects, writing of a scientific paper), computing cluster (HPC)

Languages : French (native), English (C1), Spanish (B2), Italian (B2)

Oral Presentations

Research talks :

- Action Dark Energy 2026, Paris *November 2026*
- UNBIAS meeting (echoIA), Utrecht, Netherlands *September 2026*
- Conference on gravitational lensing, Sesto, Italy (presentation of a flash talk and a poster) *July 2026*
- UNIONS meeting, Paris *July 2026*
- Intrinsic alignment workshop, IAP, Paris *June 2026*
- Elbereth conference, Paris *April 2026*
- Deep CosmoStat Days, Saclay *February 2025*
- Euclid-France, 14th conference, Strasbourg *November 2025*
- UNIONS meeting, Paris *October 2025*
- WL-SHE Euclid meeting, Marseille (France)
- UNIONS meeting, Leiden (Netherlands), talk on my post-master internship *March 2025*
- Meeting with IRAP cosmology group, Pic du Midi, talk on my M2 internship *July 2024*
- IRAP internship presentations, talk on my M2 internship *May 2024*
- *My internship in 180 seconds*, IJCLab, talk on my L3 internship *July 2022*

Conferences (attendee) :

- WL-SHE Euclid meeting, Naples (Italie) *October 2026*
- Euclid consortium meeting, Barcelona *May 2026*
- Journées des Jeunes Chercheurs, CNES *April 2026*
- Euclid-France, 13th symposium, CEA Paris-Saclay *January 2025*
- Euclid summer school "Rodolphe Clédassou" (Hendaye, Montricher-Albanne, Pornichet), student in cycle 1, cycle 2 obs and cycle 2 th (presentation of a poster in cycle 2 obs and th) *August 2024, 2025, 2026 (6 weeks)*

- Meeting on the combination of cosmological data from galaxy surveys, Institut d'Astrophysique de Paris — *June 2024*
- National CMB-France symposium, Institut d'Astrophysique de Paris — *June 2022*

Science outreach:
- Conference for ALCOR members, 15-minute presentation on my L3 internship — *December 2022*

Other professional experiences

Teaching

TD teacher during the PhD :
- Electromagnetism, 1st of Ecole d'ingénieur Denis Diderot — *January-April 2026*
- Statistical physics, Licence 3 of fundamental physics, Université Paris Cité

Private tutoring in physics, mathematics, and chemistry for final-year high school students — *2020 – 2023*

Responsabilities for the laboratory

Member of the seminar organization commitee of the Department of Astrophysics of CEA — *2026*

Personal Involvement

Member of the French Astronomy Association (AFA) — *2024 - 2025*
Animator in astronomy during "Paris sous les étoiles" (from June to August 2025)
Volunteer at the Rencontres du Ciel et de l'Espace (November 2024)
Animator "Débusquez les infox" in middle schools and libraries

Vice-secretary of the ALCOR association (Astronomy and Lights of the Orsay Campus) — *2023 - 2024*
Volunteer with the association at the Fleurance Astronomy Festival (August 2022, 2023, 2024 and 2025)

Member of a collective for the Jacques Tati cinema in Orsay — *2023 - 2024*
Organization of film debates

Cultural coordinator of the "Photon" student association — *2022 - 2023*
Organization of cultural outings in Paris

Interests

Music: classical guitar solo and ensemble (8 years at Le Mans Conservatory, awarded the Music Studies Certificate in 2019), folk guitar (member of the music band Puls'art)

Sports: competitive tennis (best ranking: 15), jogging, climbing, surfing, trail running

Travel: London, United States (Boston, Chicago, Los Angeles, New York), Spain (Seville), Italy, Sardinia, Sicily, Scotland (Edinburgh), Belgium (Bruxelles)

Other hobbies: reading, writing, philosophy, drawing

References

Martin Kilbinger, CEA Astrophysics Department, CosmoStat team, Saclay (current post-master internship supervisor)
- email: martin.kilbinger@cea.fr

Isaac Tutusaus, Institut de Recherche en Astrophysique et Planétologie (IRAP), Toulouse (M2 internship supervisor)
- email: isaac.tutusaus@irap.omp.eu

Marian Douspis, Institut d'Astrophysique Spatiale (IAS), Orsay (M2 numerical project supervisor)
- email: marian.douspis@universite-paris-saclay.fr

Mathieu Vincendon, Institut d'Astrophysique Spatiale (IAS), Orsay (M2 director)
- email: mathieu.vincendon@universite-paris-saclay.fr